

# CHE 565 -- Homework 2

Soki Sem

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## Problem 1

### Setup

$$\begin{aligned}\text{Option A: } C_{0,A} &= 3,800,000 \\ FV_A &= 1,100,000/\text{yr} \\ \text{Option B: } C_{0,B} &= 5,000,000 \\ FV_B &= 1,410,000/\text{yr}\end{aligned}$$

$$\begin{aligned}NPV &= PV + C_0 \\ PV &= F \left[ \frac{(1+r)^n - 1}{r(1+r)^n} \right]\end{aligned}$$

PV = present value, FV = future value (annual cash flow), r = yearly interest rate, n = number of years  
n = 10 years, r = 0.10

$$NPV(A) = \$2,959,023.82, \quad NPV(B) = \$3,663,839.62$$

NPV is higher for option B at 10% yearly interest, so B is preferred under these assumptions.

b.) 10 year lifetime, no salvage value, and 5 % yearly interest rate. What will be the yearly payment?

$$P = C_0 * \frac{(1+r)^nr}{(1+r)^n - 1}$$

P = annual payment, C0 = initial cost, r = yearly interest rate, n = number of years

$$P(A) = \$ - 492,117.38/\text{year}, \quad P(B) = \$ - 647,522.87/\text{year}$$

## Problem 2

### Setup

\$0.94/lbs is the price of powdered detergent.

0.1 % of 4e6 lbs of detergent per year is carried through the exhaust

Adding a second cyclone separator costs \$10,000, with \$300/year in maintenance costs will reduce carryover to 0.0%, with a 8% yearly interest rate.

Cost of carryover detergent: \$3,760.00/year

Cost of second separator: \$10,000.00 initial + \$600.00/year maintenance = \$10,600.00 total

## Problem 2

### Setup

\$0.94/lb is the price of powdered detergent.

0.1% of 4e6 lb/year is carried through the exhaust.

Adding a second cyclone separator costs \$10,000 with \$300/year maintenance and reduces carryover to 0.0%.

Lost detergent value: \$3,760.00/year

a.) Find the additional yearly income

Gross additional income (recovered detergent): \$3,760.00/year

Net additional income after maintenance: \$3,460.00/year

b.) Find the payback period

Payback period: 2.89 years (34.7 months)

## Problem 3

### Setup

Determine if each function is convex

$$A : f(x) = (x_1 - x_2)^2 + x_2^2$$

$$B : f(x) = x_1^2 + x_2^2 + x_3^2$$

$$C : f(x) = \exp(x_1) + \exp(x_2)$$

$$A: 1 \leq \text{im}[1 \ 1]$$